NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2023-2024

EE7403 – IMAGE ANALYSIS & PATTERN RECOGNITION

April / May 2024 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. An edge detection algorithm produces a binary image:

$$f(x,y)=\delta(x,y-1)+\delta(x-1,y-2)+\delta(x+1,y-2)+\delta(x-2,y-3)+\delta(x+2,y-3).$$

The possible pixels (x, y) on the edge are indicated by $f(x,y)>0$. A Hough transform $g(a,b)$ of model $y=ax+b$ is applied to detect the lines in $f(x,y)$.

(a) What are all the possible values of this Hough transform $g(a,b)$?

(5 Marks)

(b) How many points in the Hough transform satisfy $g(a,b)>1$?

(5 Marks)

(c) Determine the equation(s) of line(s) $y=ax+b$ that has/have the highest number of pixels.

(7 Marks)

(d) Let

$$G(a,b)=\begin{cases} g(a,b), & \text{if} \quad g(a,b)>1 \\ 0, & \text{otherwise} \end{cases}.$$

Determine $G(a,b)$.

(8 Marks)

2. Suppose that the class prior probability and the class-conditional probability density function (PDF) are given as $p(\omega_i)$ and $p(\mathbf{x} \mid \omega_i)$ for different classes ω_i, $i = 1, 2, \ldots, c$.

   (a) Express the theoretical error rate of classifying an input sample $\mathbf{x}$ to class ω_i in terms of $p(\omega_i)$ and $p(\mathbf{x} \mid \omega_i)$ for $\omega_i, i = 1, 2, \ldots, c$.

   (8 Marks)

   (b) From part (a), derive and simplify the decision rule that minimizes the theoretical error rate.

   (5 Marks)

   (c) Explain how to use training samples to estimate $p(\omega_i)$, $p(\mathbf{x} \mid \omega_i)$ and $p(\mathbf{x})$ so that the decision rule that minimizes the error rate leads to the simple k-nearest neighbor classifier.

   (12 Marks)

3. A dataset contains 5 two-dimensional samples: $\mathbf{x}_1 = [5 \quad 1]^T$, $\mathbf{x}_2 = [4 \quad 2]^T$, $\mathbf{x}_3 = [3 \quad 3]^T$, $\mathbf{x}_4 = [2 \quad 4]^T$ and $\mathbf{x}_5 = [1 \quad 5]^T$.

   (a) Compute the sample covariance matrix of the dataset.

   (7 Marks)

   (b) Plot the five samples as five points in the data space $\mathbf{x}$.

   (3 Marks)

   (c) Obtain the eigenvalues and the unit-length eigenvectors of the covariance matrix obtained in part (a).

   (8 Marks)

   (d) Explain the interpretation of the eigenvalues and the eigenvectors of a covariance matrix for the results in parts (a), (b) and (c).

   (7 Marks)

4. In an image application, the input feature maps of spatial size $P \times Q$ with C channels and the output feature maps of spatial size $P \times Q$ with D channels of a layer of neural network are expressed as $x_{i,j,k}, 1 \le i \le P, 1 \le j \le Q, 1 \le k \le C$, and $y_{i,j,k}, 1 \le i \le P, 1 \le j \le Q, 1 \le k \le D$, respectively.

   (a) Express the output feature maps $y_{i,j,k}$ in terms of the input feature maps $x_{i,j,k}$ via the scalar network parameters of a layer of fully connected neural network. What is the number of learnable parameters?

   (5 Marks)

Note: Question No. 4 continues on page 3.

(b) Express the output feature maps in terms of the input feature maps via the scalar network parameters of a layer of spatial convolutional neural network of filter size 3×3. What is the number of learnable parameters?

(5 Marks)

(c) Express the output feature maps in terms of the input feature maps via the scalar network parameters of a layer of spatial convolutional neural network of filter size 1×1. What is the number of learnable parameters?

(5 Marks)

(d) If we use a single index for the 2D spatial position to express the input and output feature maps by $x_{i,k}, 1 \le i \le PQ, 1 \le k \le C$, and $y_{i,k}, 1 \le i \le PQ, 1 \le k \le D$, respectively, re-express the answer to part (c) without using the bias. What is the number of learnable parameters?

(5 Marks)

(e) Arrange all $x_{i,k}$ in part (d) into a $PQ \times C$ matrix, $\boldsymbol{X}$, and arrange all $y_{i,k}$ in part (d) into a $PQ \times D$ matrix, $\boldsymbol{Y}$. Re-express the answer to part (d) in the matrix format $\boldsymbol{X}$ and $\boldsymbol{Y}$. (Note that $PQ \times C$ matrix has PQ rows and C columns.) What can you conclude about the relation between a convolutional neural network and a Transformer?

(5 Marks)

[**Hint for Question 4:** If inputs and outputs of a layer of network are expressed as $x_i, 1 \le i \le P$ and $y_i, 1 \le i \le Q$, respectively, the outputs of a layer of fully connected neural network can be expressed in terms of the inputs as $y_i = \sum_{l=1}^{P} w_{l,i} x_l + b_i, 1 \le i \le Q$, where $w_{l,i}, b_i$ are the scalar parameters called the weights and biases of the network, respectively.]

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.